

Project Design Phase

Solution Architecture

Date	06 July 2026
Team ID	Individual
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Project Name	India's Agricultural Crop Production Analysis (1997–2021)
Maximum Marks	4 Marks

Solution Architecture:

The proposed system for the Agricultural Crop Production Analysis follows a simple data analytics workflow using Tableau for visualization and data processing. The architecture starts with collecting raw crop production data, followed by data cleaning and preparation. The cleaned data is then imported into Tableau for analysis and visualization. Finally, interactive dashboards are created to generate meaningful insights about crop production trends across states, crops, and seasons over the period 1997–2021.

System Flow:



- **Data Source (CSV Dataset)** – Crop production data such as state, district, crop, year, season, area cultivated, and production quantity.
- **Data Preprocessing (Excel / Data Cleaning)** – Removing missing values, correcting formats, and organizing data.
- **Data Visualization Tool (Tableau)** – Import cleaned dataset and create charts.
- **Dashboard Creation** – State-wise, crop-wise, season-wise, and year-wise production analysis.
- **Insights & Decision Making** – Identifying high-yield crops, regional trends, and production patterns over the years.

Conclusion:

This architecture helps in transforming raw agricultural data into meaningful insights using visualization techniques, supporting better understanding of crop production trends across India from 1997 to 2021.

Description:

The Agricultural Crop Production dataset (CSV) is collected as the data source, then cleaned during the data preparation stage (removing duplicates, handling missing values, and correcting formats). The cleaned data is imported into Tableau for analysis, where state-wise, crop-wise, season-wise, area vs production, and yield-based visualizations are created. These visualizations are combined into an interactive dashboard and then organized into a Tableau Story that presents the analysis step by step. Finally, insights and recommendations on crop production trends are generated as the project output.